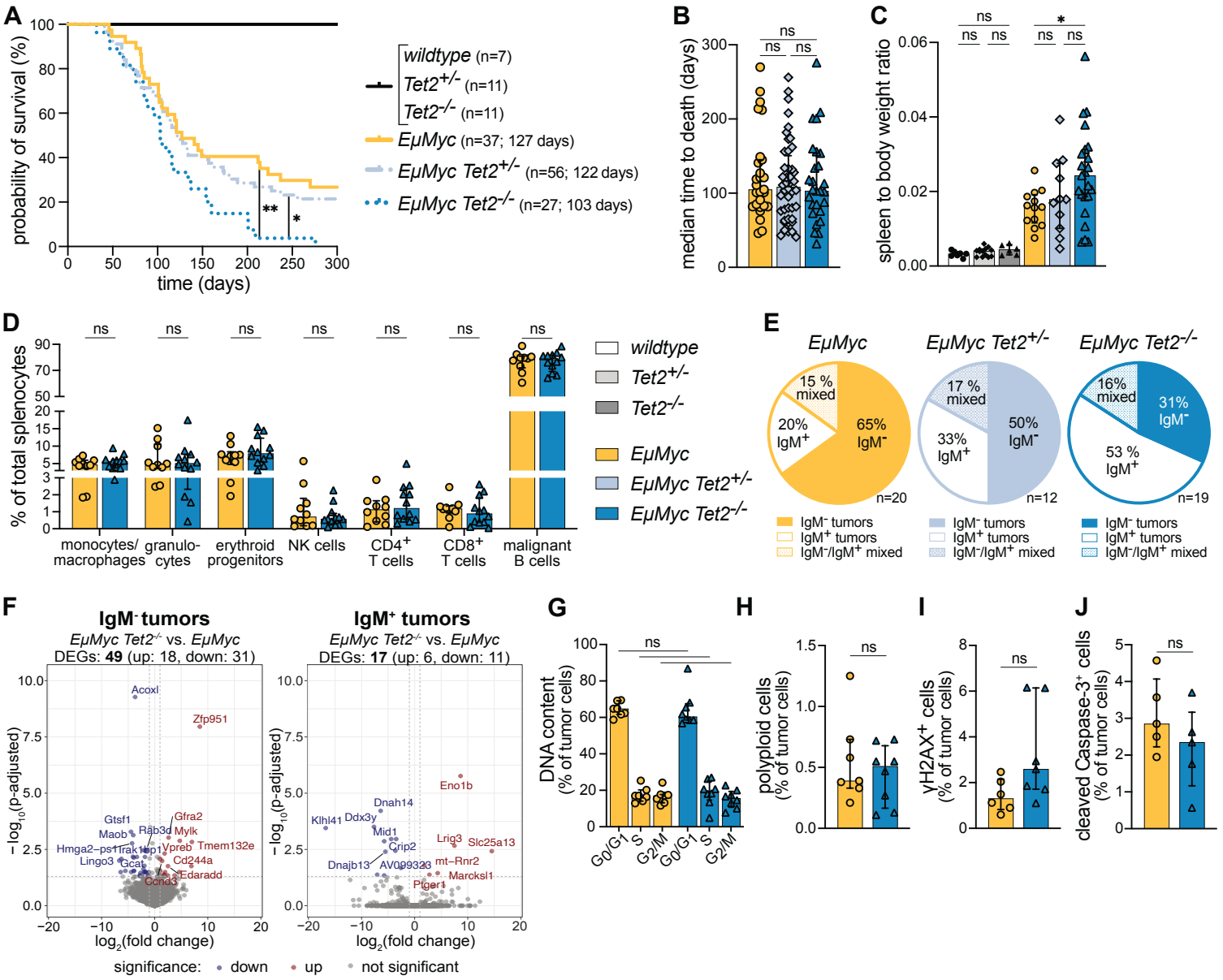
FIGURE 1


A
probability of survival (%)
time (days)
wildtype (n=7)
Tet2+/- (n=11)
Tet2-/- (n=11)
EμMyc (n=37; 127 days)
EμMyc Tet2+/- (n=56; 122 days)
EμMyc Tet2-/- (n=27; 103 days)
B
median time to death (days)
C
spleen to body weight ratio
D
% of total splenocytes
monocytes/ macrophages
granulo- cytes
erythroid progenitors
NK cells
CD4+ T cells
CD8+ T cells
malignant B cells
E
EμMyc
EμMyc Tet2+/-
EμMyc Tet2-/-
15 % mixed
20% IgM+
65% IgM-
n=20
17 % mixed
33% IgM+
50% IgM-
n=12
16% mixed
31% IgM-
53 % IgM+
n=19
IgM- tumors
IgM+ tumors
IgM-/IgM+ mixed
F
IgM- tumors
EμMyc Tet2-/- vs. EμMyc
DEGs: 49 (up: 18, down: 31)
IgM+ tumors
EμMyc Tet2-/- vs. EμMyc
DEGs: 17 (up: 6, down: 11)
log2(fold change)
significance: down up not significant
G
DNA content (% of tumor cells)
H
polyploid cells (% of tumor cells)
I
γH2AX+ cells (% of tumor cells)
J
cleaved Caspase-3+ cells (% of tumor cells)

Supplementary Figure 1

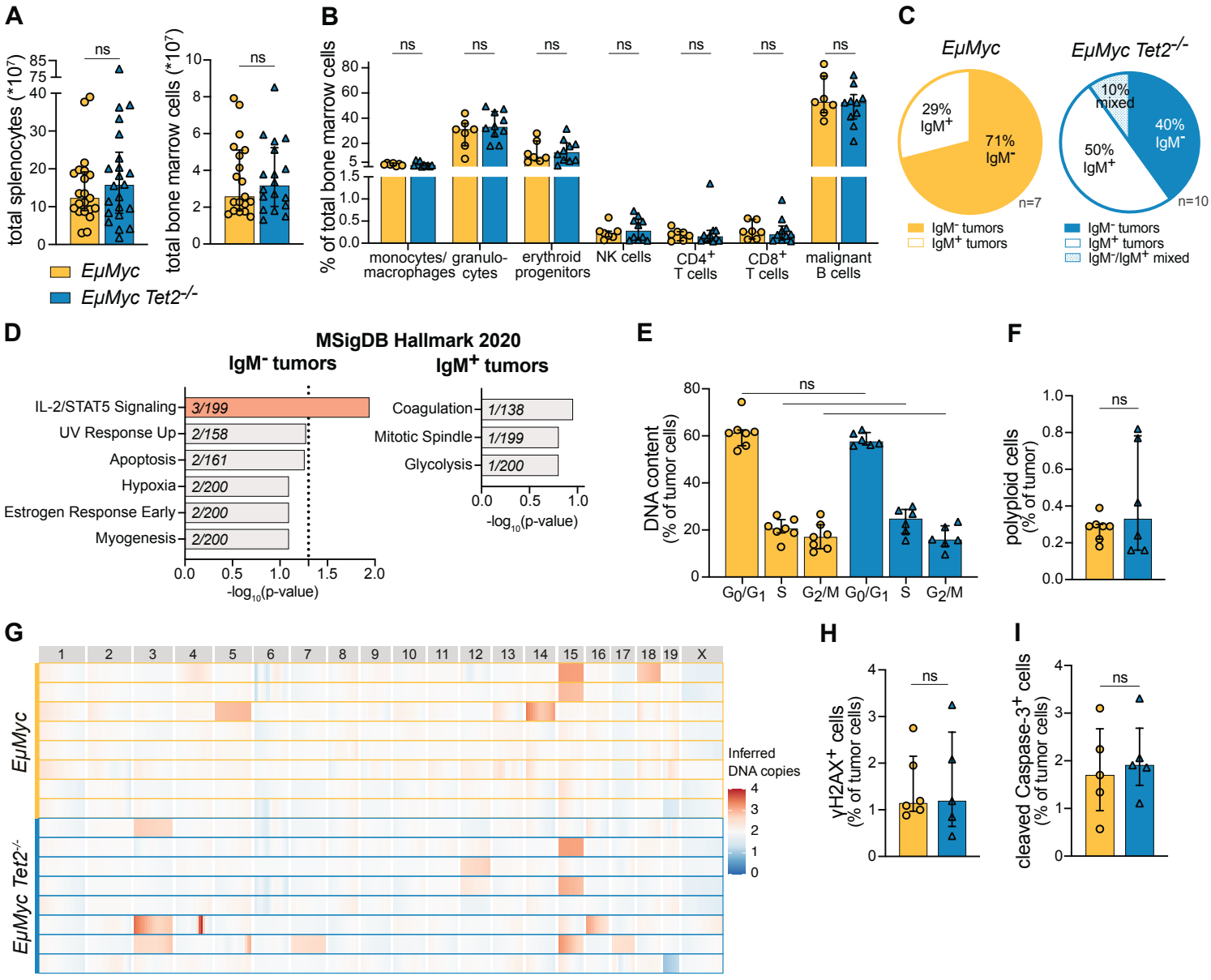
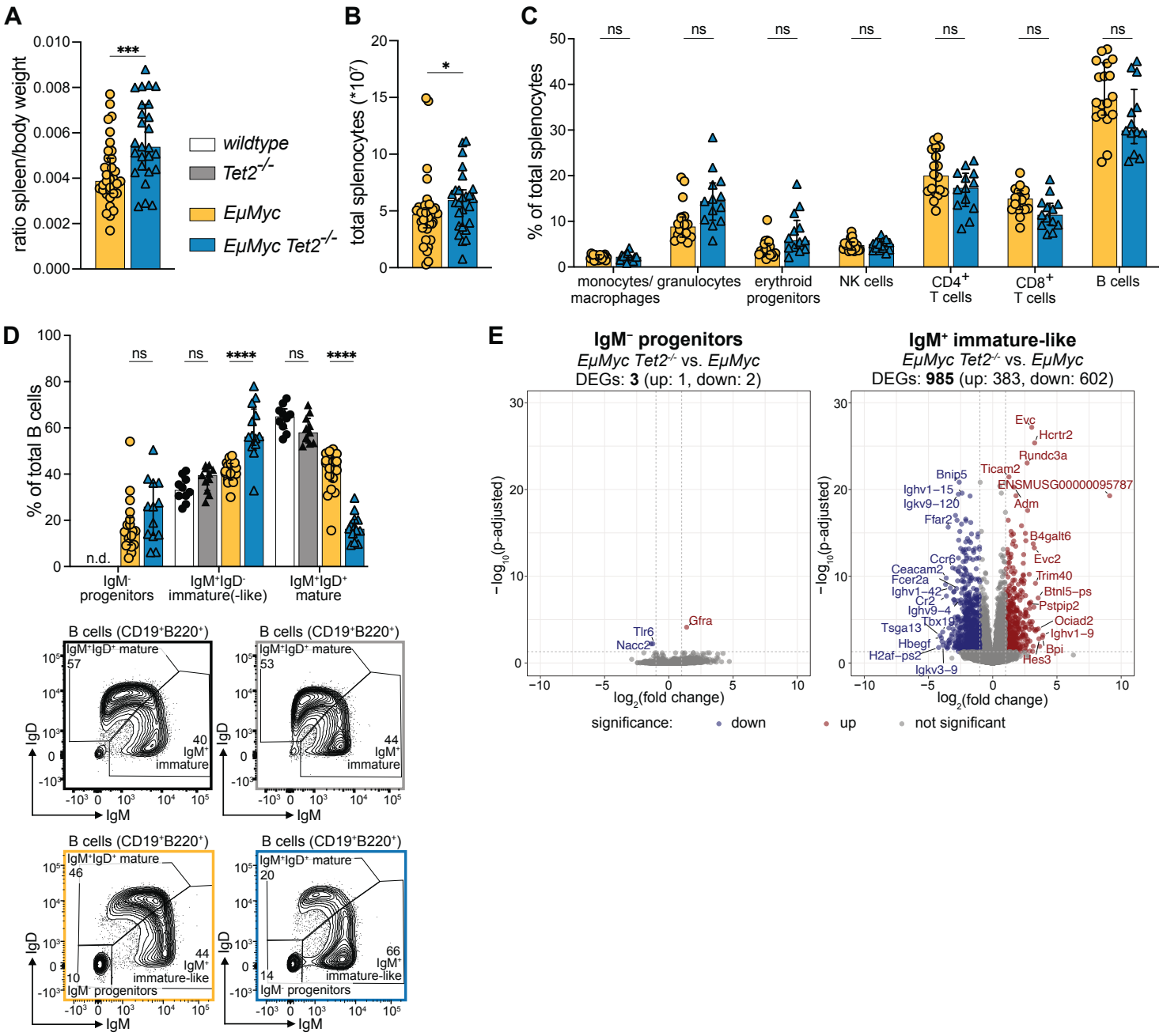


FIGURE 2


A
ratio spleen/body weight
***
wildtype
Tet2-/-
EμMyc
EμMyc Tet2-/-
B
total splenocytes (*10^7)
*
C
% of total splenocytes
ns
monocytes/ macrophages
granulocytes
erythroid progenitors
NK cells
CD4+ T cells
CD8+ T cells
B cells
D
% of total B cells
n.d.
****
IgM- progenitors
IgM+IgD- immature(-like)
IgM+IgD+ mature
B cells (CD19+B220+)
IgM+IgD+ mature
57
40
IgM+ immature
53
44
46
20
66
immature-like
IgM- progenitors
10
14
IgD
IgM
E
IgM- progenitors
EμMyc Tet2-/- vs. EμMyc
DEGs: 3 (up: 1, down: 2)
IgM+ immature-like
DEGs: 985 (up: 383, down: 602)
Gfra
Tlr6
Nacc2
Evc
Hcrtr2
Rundc3a
Ticam2
Bnip5
Ighv1-15
Igkv9-120
ENSMUSG00000095787
Adm
Ffar2
B4galt6
Evc2
Ccr6
Ceacam2
Fcer2a
Ighv1-42
Trim40
Btnl5-ps
Cr2
Ighv9-4
Pstpip2
Tbx19
Tsga13
Ociad2
Ighv1-9
Hbegf
Bpi
H2af-ps2
Hes3
Igkv3-9
-log10(p-adjusted)
log2(fold change)
significance:
down
up
not significant

Supplementary Figure 2

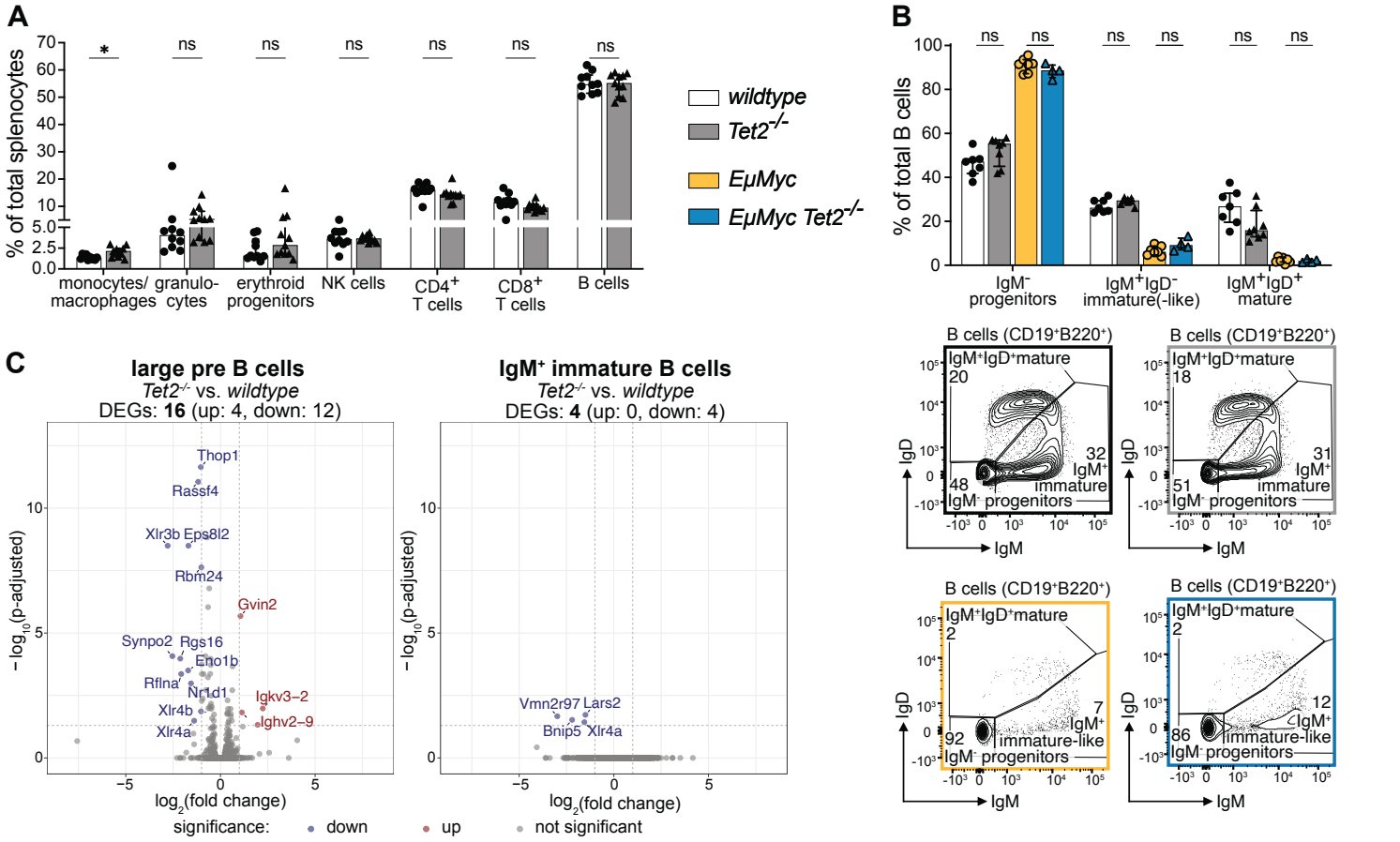
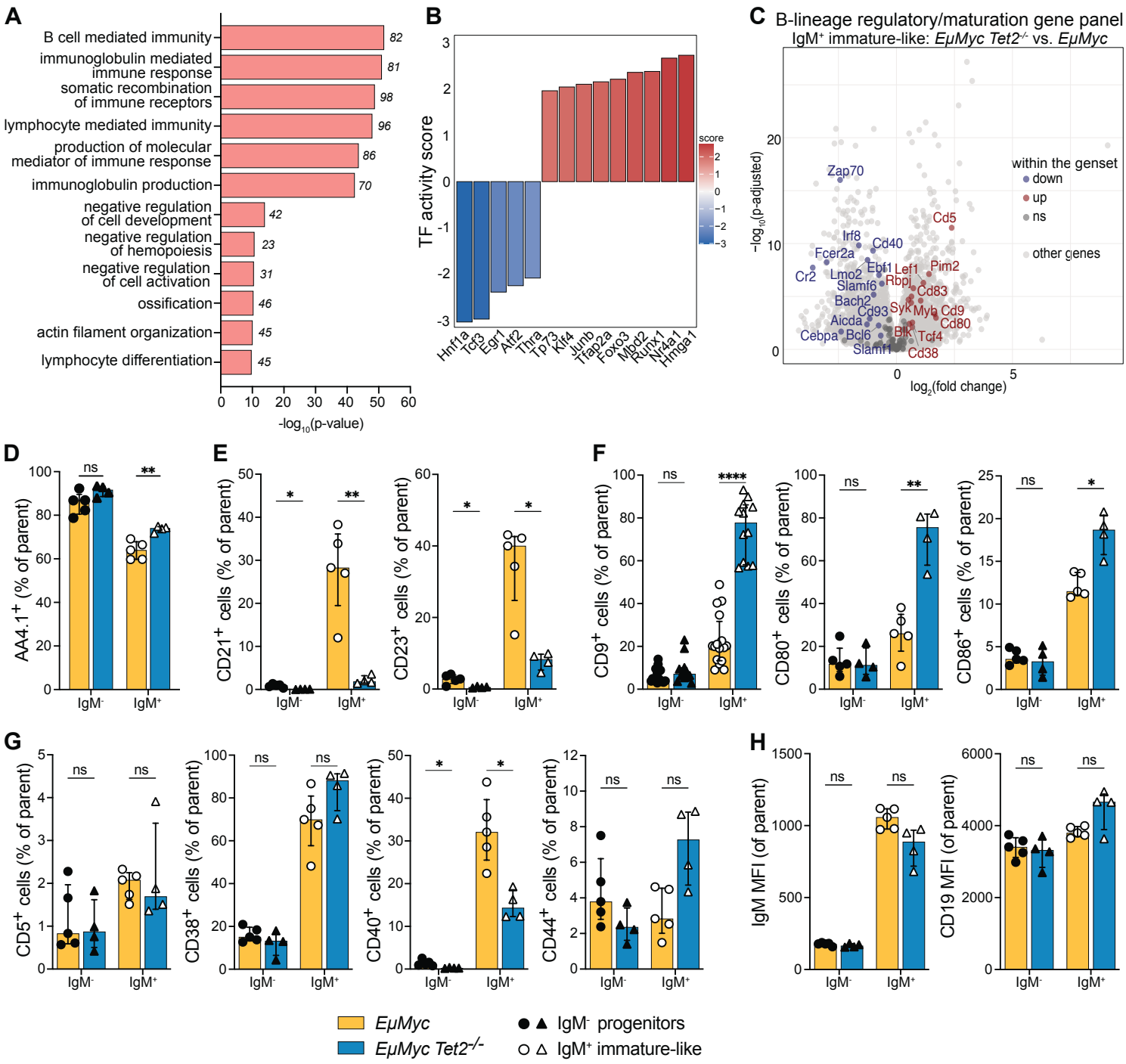


FIGURE 3


A
B cell mediated immunity
immunoglobulin mediated immune response
somatic recombination of immune receptors
lymphocyte mediated immunity
production of molecular mediator of immune response
immunoglobulin production
negative regulation of cell development
negative regulation of hemopoiesis
negative regulation of cell activation
ossification
actin filament organization
lymphocyte differentiation
82
81
98
96
86
70
42
23
31
46
45
45
-log10(p-value)
B
TF activity score
score
Hnf1a
Tcf3
Egr1
Atf2
Thra
Tp73
Klf4
Junb
Tfap2a
Foxo3
Mpd2
Runx1
Nr4a1
Hmga1
C
B-lineage regulatory/maturation gene panel
IgM+ immature-like: EμMyc Tet2-/- vs. EμMyc
within the genset
down
up
ns
other genes
Zap70
Irf8
Fcer2a
Cr2
Cd40
Ebf1
Lef1
Pim2
Lmo2
Slamf6
Rbpj
Cd5
Cd83
Bach2
Cd93
Syk
Myb
Cd9
Aicda
Cd80
Cebpa
Bcl6
Blk
Tcf4
Slamf1
Cd38
−log10(p-adjusted)
log2(fold change)
D
AA4.1+ (% of parent)
E
CD21+ cells (% of parent)
CD23+ cells (% of parent)
F
CD9+ cells (% of parent)
CD80+ cells (% of parent)
CD86+ cells (% of parent)
G
CD5+ cells (% of parent)
CD38+ cells (% of parent)
CD40+ cells (% of parent)
CD44+ cells (% of parent)
H
IgM MFI (of parent)
CD19 MFI (of parent)
ns
IgM-
IgM+
EμMyc
EμMyc Tet2-/-
IgM- progenitors
IgM+ immature-like

Supplementary Figure 3

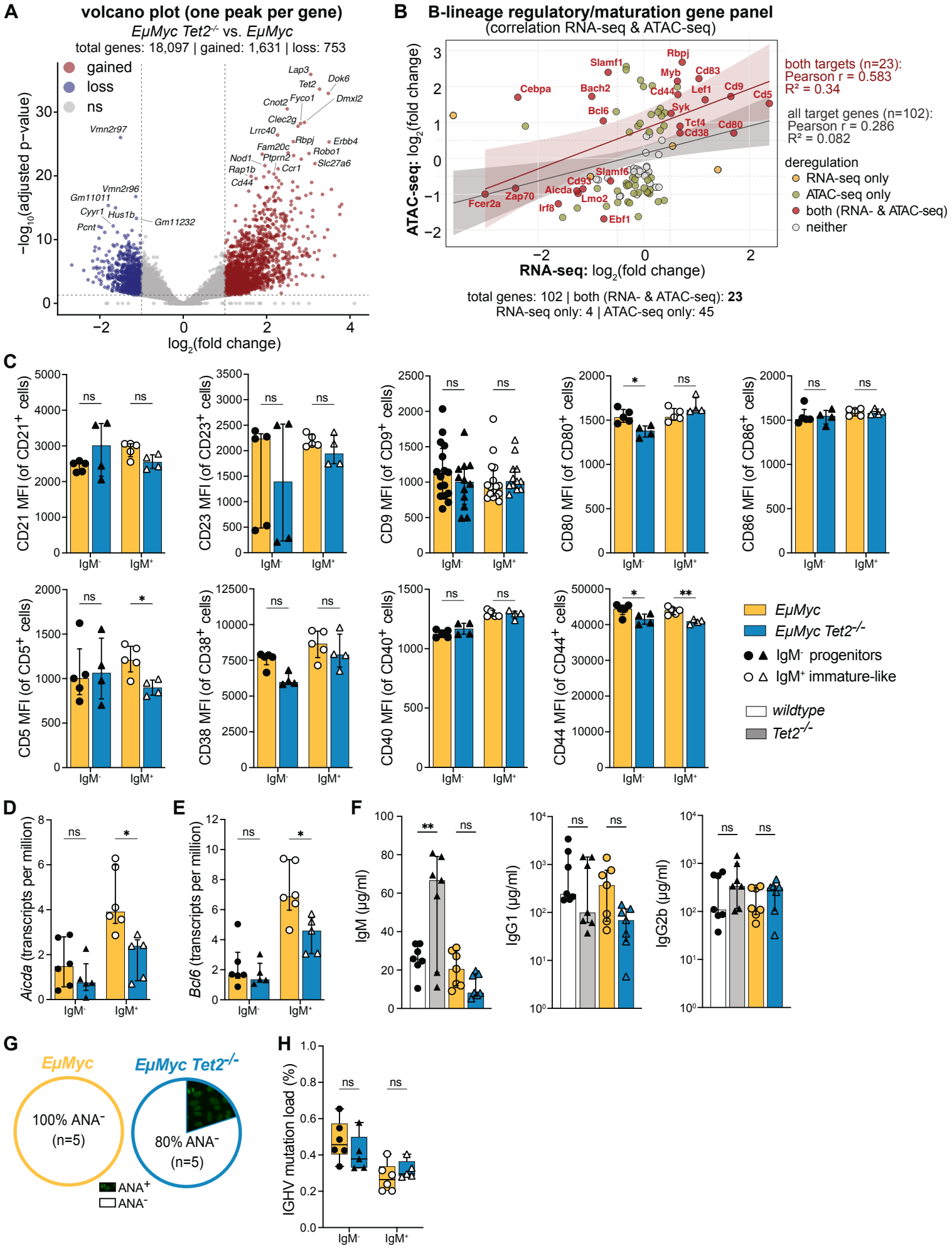
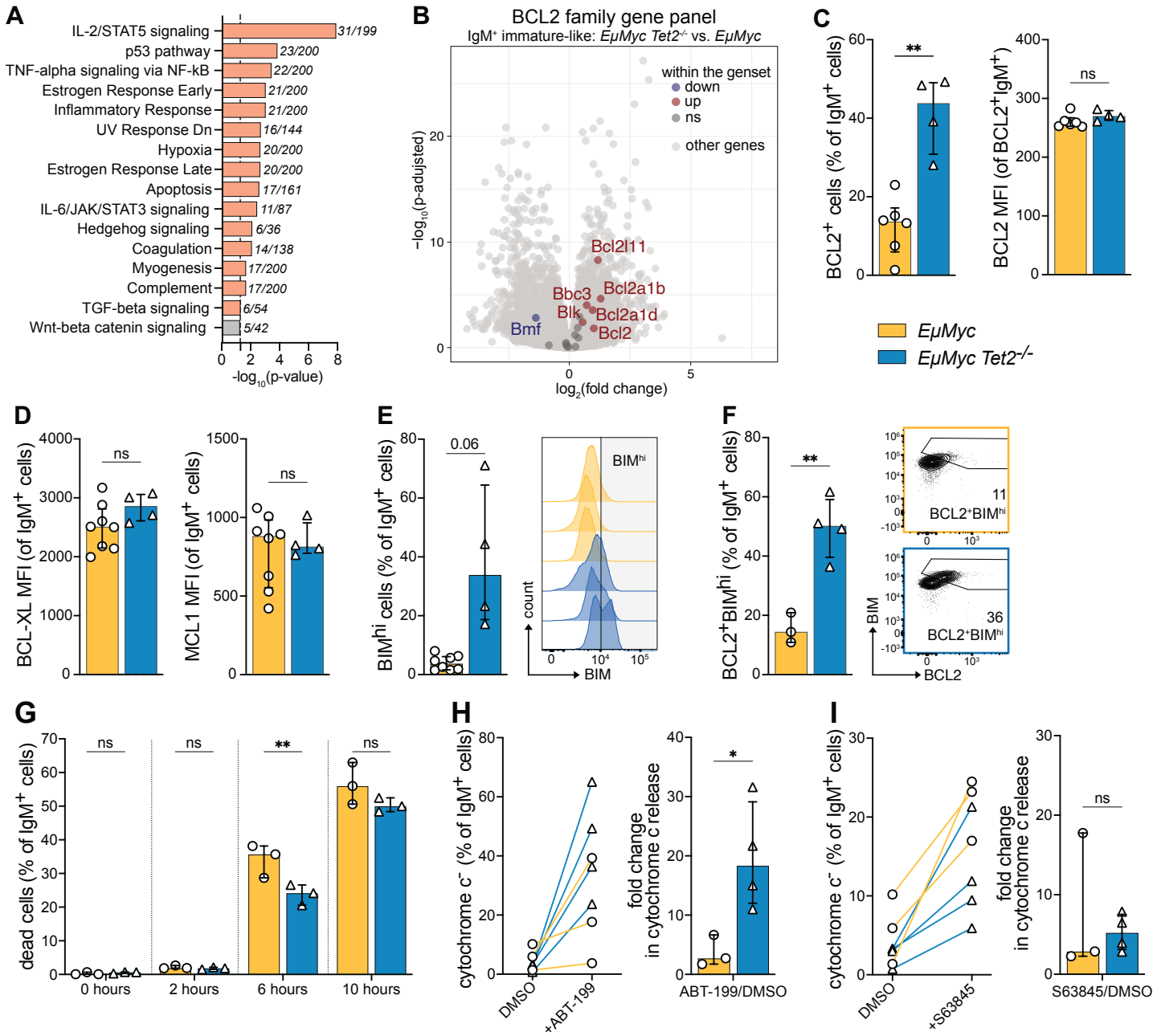
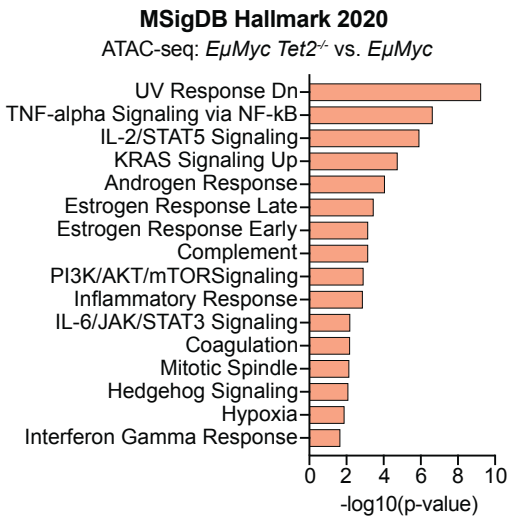


FIGURE 4


A
IL-2/STAT5 signaling 31/199
p53 pathway 23/200
TNF-alpha signaling via NF-kB 22/200
Estrogen Response Early 21/200
Inflammatory Response 21/200
UV Response Dn 16/144
Hypoxia 20/200
Estrogen Response Late 20/200
Apoptosis 17/161
IL-6/JAK/STAT3 signaling 11/87
Hedgehog signaling 6/36
Coagulation 14/138
Myogenesis 17/200
Complement 17/200
TGF-beta signaling 6/54
Wnt-beta catenin signaling 5/42
-log10(p-value)
B
BCL2 family gene panel
IgM+ immature-like: EμMyc Tet2-/- vs. EμMyc
within the genset
down
up
ns
other genes
Bcl2l11
Bbc3
Bcl2a1b
Blk
Bcl2a1d
Bmf
Bcl2
-log10(p-adjusted)
log2(fold change)
C
BCL2+ cells (% of IgM+ cells)
BCL2 MFI (of BCL2+IgM+)
ns
EμMyc
EμMyc Tet2-/-
D
BCL-XL MFI (of IgM+ cells)
MCL1 MFI (of IgM+ cells)
ns
E
BIMhi cells (% of IgM+ cells)
0.06
BIMhi
count
BIM
F
BCL2+BIMhi (% of IgM+ cells)
11
BCL2+BIMhi
36
BCL2+BIMhi
BIM
BCL2
G
dead cells (% of IgM+ cells)
ns
ns
**
ns
0 hours
2 hours
6 hours
10 hours
H
cytochrome c- (% of IgM+ cells)
DMSO
+ABT-199
fold change in cytochrome c release
ABT-199/DMSO
I
cytochrome c- (% of IgM+ cells)
DMSO
+S63845
fold change in cytochrome c release
S63845/DMSO

Supplementary Figure 4



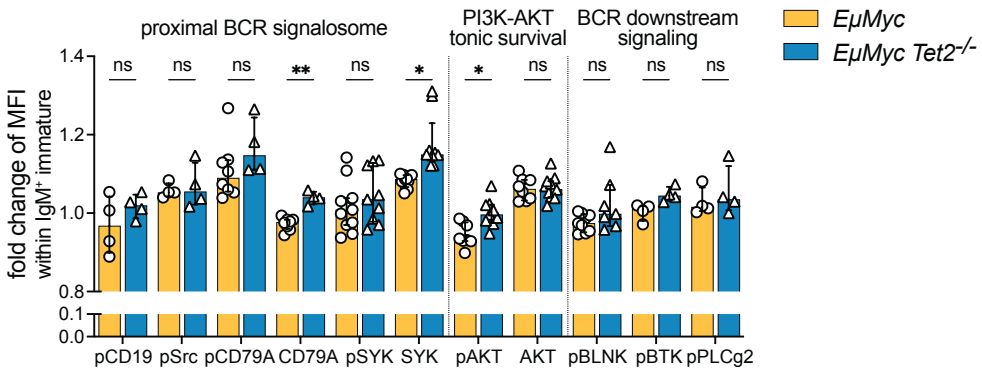
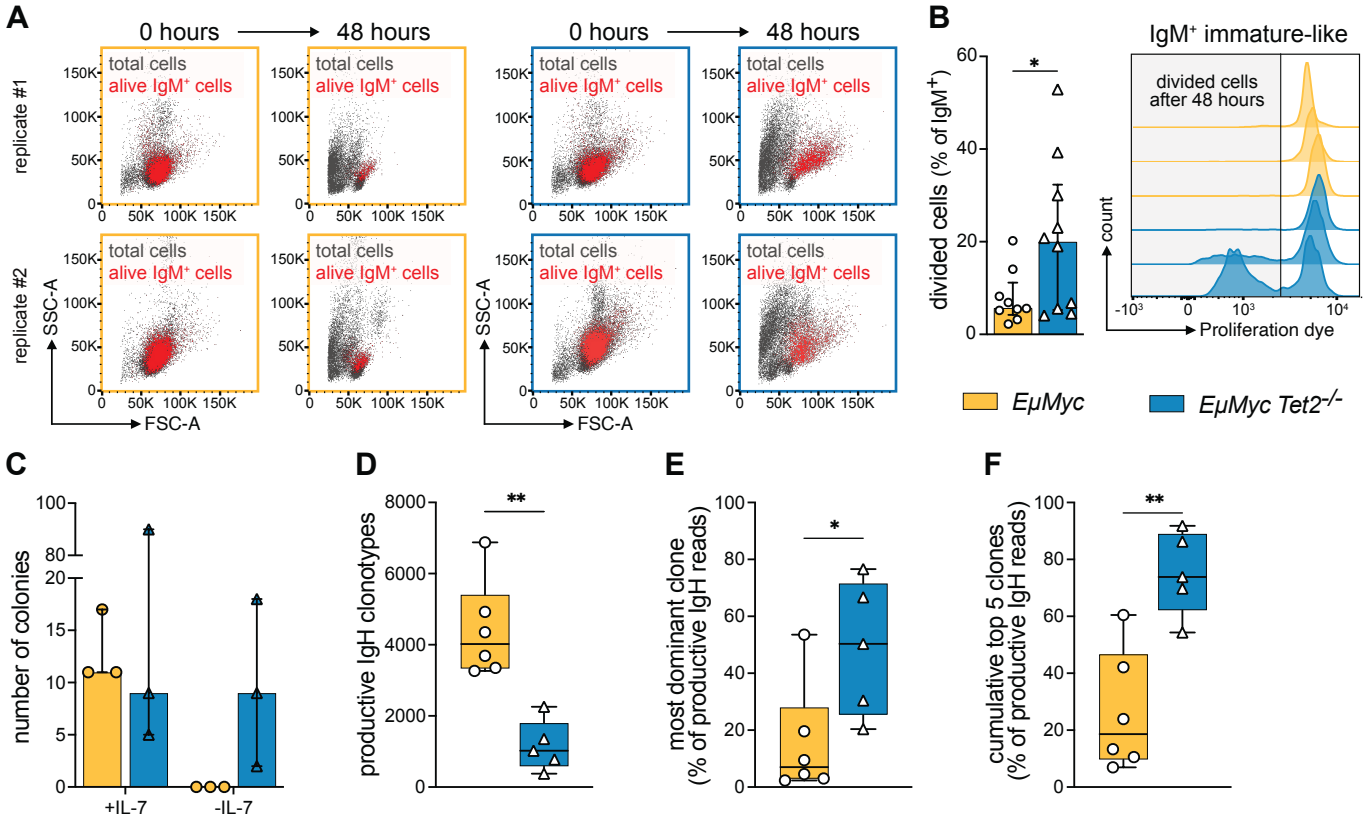


FIGURE 5


A
0 hours
48 hours
0 hours
48 hours
replicate #1
replicate #2
total cells
alive IgM+ cells
SSC-A
FSC-A
B
IgM+ immature-like
divided cells (% of IgM+)
divided cells after 48 hours
count
Proliferation dye
EμMyc
EμMyc Tet2-/-
C
number of colonies
+IL-7
-IL-7
D
productive IgH clonotypes
E
most dominant clone (% of productive IgH reads)
F
cumulative top 5 clones (% of productive IgH reads)

Supplementary Figure 5

